Verification of Thevenin’s Theorem	
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Aim:

Verification of Thevenin’s Theorem:

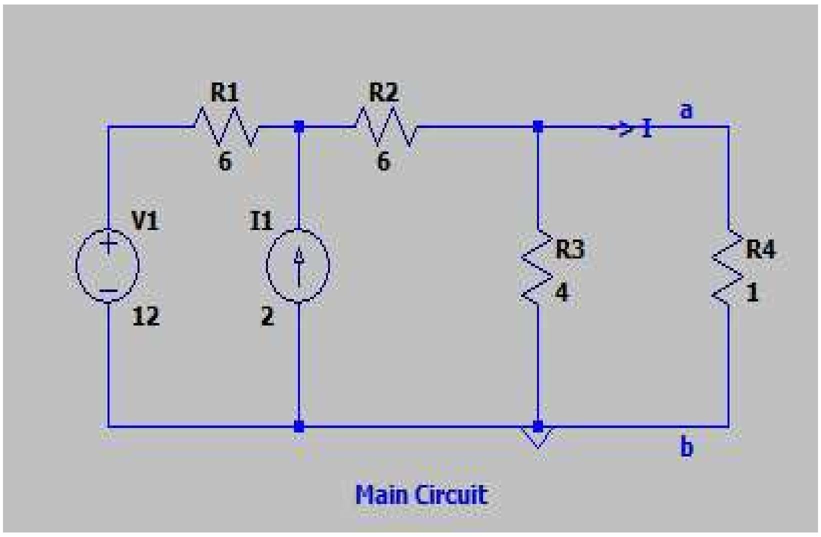
Theory:

Thevenin’s theorem states that any linear circuit containing several voltage sources and resistors can be simplified to a Thevenin-equivalent circuit with a single voltage source and resistance connected in series with a load.

Question 1

Using Thevenin's theorem find the equivalent circuit to the left of the terminals in the circuit of Fig. then find I

Circuit Diagram:

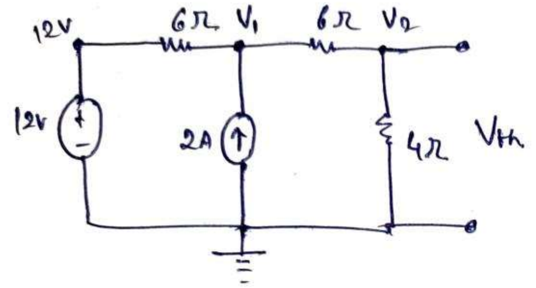


Theoretical Calculation:

Determination of Thevenin’s Voltage V_{th}:

Que:

For V_{th}.



$V_{4\Omega} = V_{th}$.

KCL at V_1

$$\frac{12-V_1}{6} + 2 = \frac{V_1-V_2}{6}.$$

$$\frac{12-V_1-V_1+V_2}{6} = -2$$

$$12-2V_1+V_2 = -12.$$

$$2V_1 - V_2 = 24 \quad \text{—} (1)$$

KCL at V_2

$$\frac{V_1-V_2}{6} = \frac{V_2}{4}$$

$$\Rightarrow 4V_1 - 4V_2 = 6V_2$$

$$\Rightarrow 4V_1 \cancel{+} 10V_2 = 0$$

$$\Rightarrow 4V_1 - 10V_2 = 0 \quad \text{—} (2)$$

$V_1 = 15$ V.

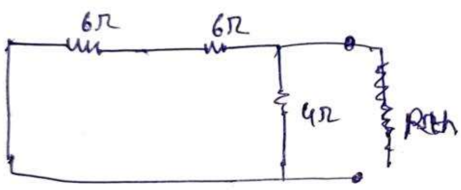
$V_2 = 6$V

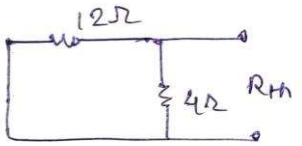
$V_{4\Omega} = 6V = V_{th}$.

$$\boxed{V_{th} = 6V}$$

Determination of Thevenin's Resistance R_{th}:

For R_{th}





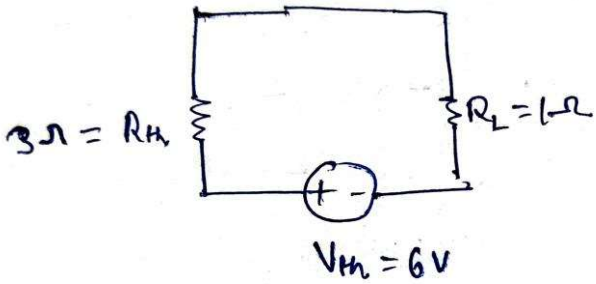
$$R_{th} = \frac{12 \times 4}{12+4}$$

$$R_{th} = \frac{\overset{3}{\cancel{12}} \times \cancel{4}}{\cancel{16}_{\cancel{4}}}$$

$$\boxed{R_{th} = 3\Omega}$$

Thevenin's Equivalent Circuit and Load Current I_L Calculation:

Que 1 Continue Thevenin's Equivalent Circuit



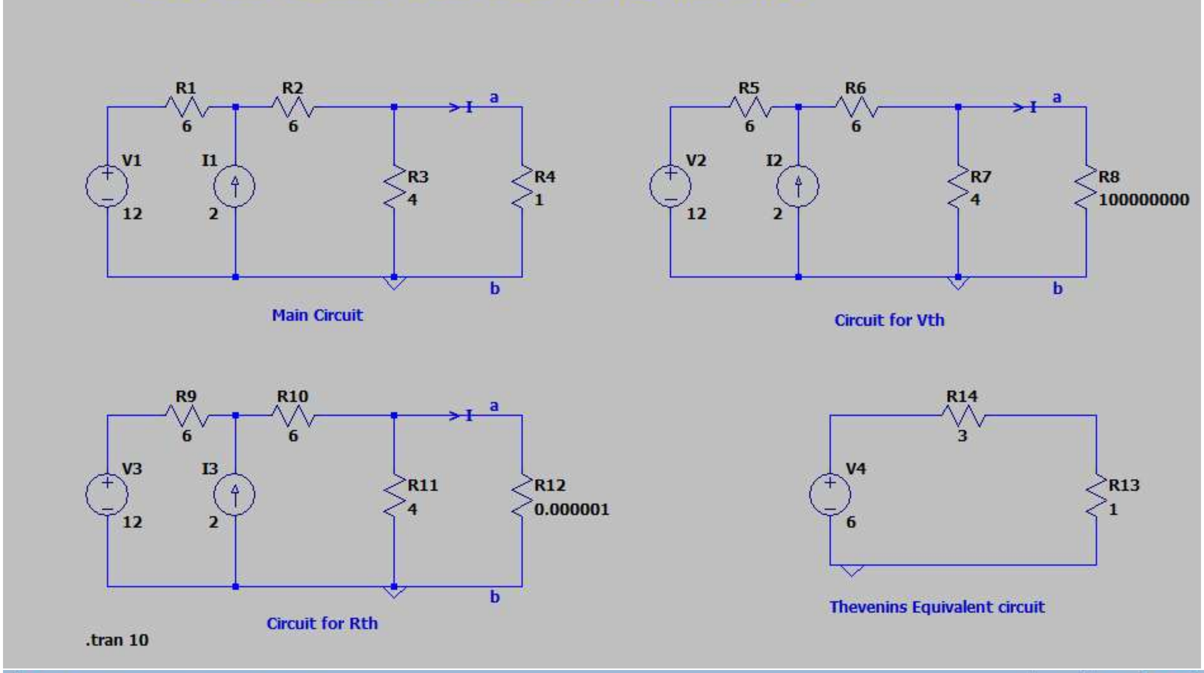
$$I_L = I_{1\Omega} = \frac{6}{4} = \frac{3}{2} = 1.5A$$

$$\boxed{I_L = 1.5A}$$

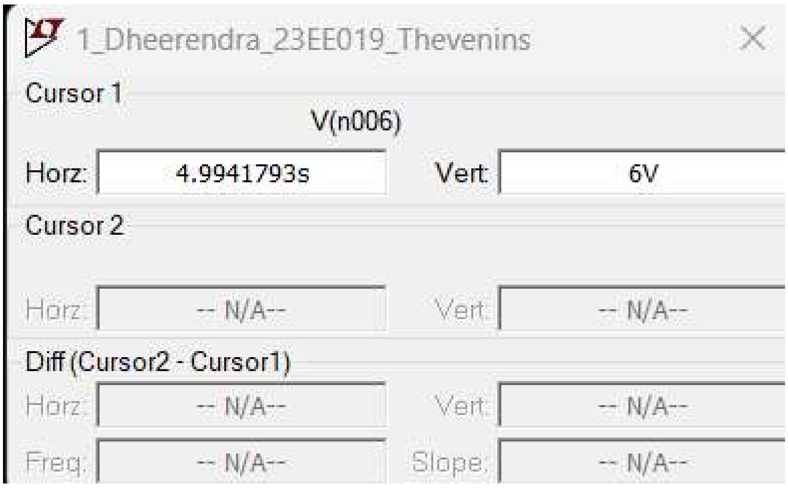
Verification by Simulation:

Circuit Diagram:

Que 1. Using Thevenin's theorem find the equivalent circuit to the left of the terminals in the circuit of Fig. Then find I



Determination of Thevenin's Voltage: 6V



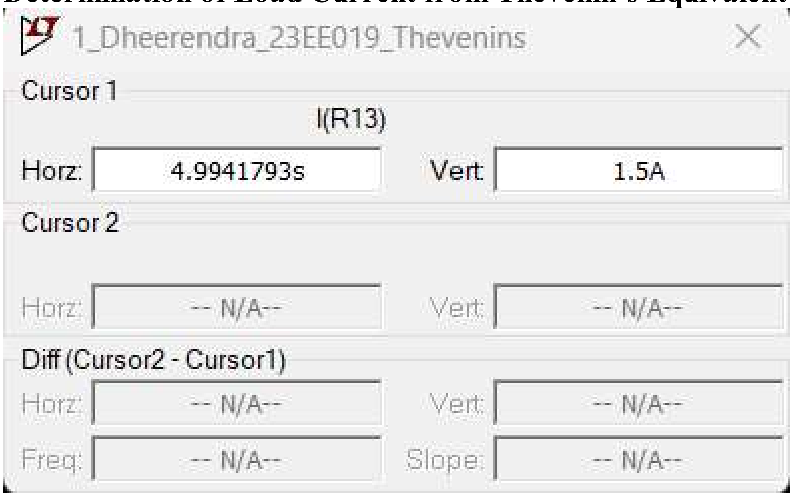
Determination of Thevenin's Resistance:

Isc =1.9999993A

Vth = 6V

Rth = Vth/Isc = 3 ohm

Determination of Load Current from Thevenin's Equivalent Circuit:



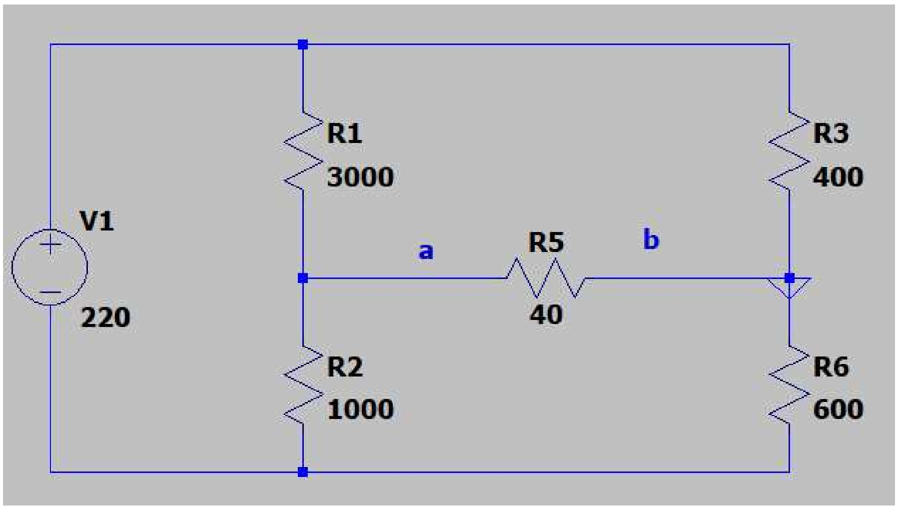
Observation Table:

S. No.	V_{th} (V)	R_{th} (Ω)	Current through Load Resistance I_L (A)			
			Practical Value		Theoretical Value	
1 (Practical)	6	3	Main Circuit	1.5	Main Circuit	1.5
1 (Theoretical)	6	3	Thevenin's Circuit	1.5	Thevenin's Circuit	1.5

Question 2

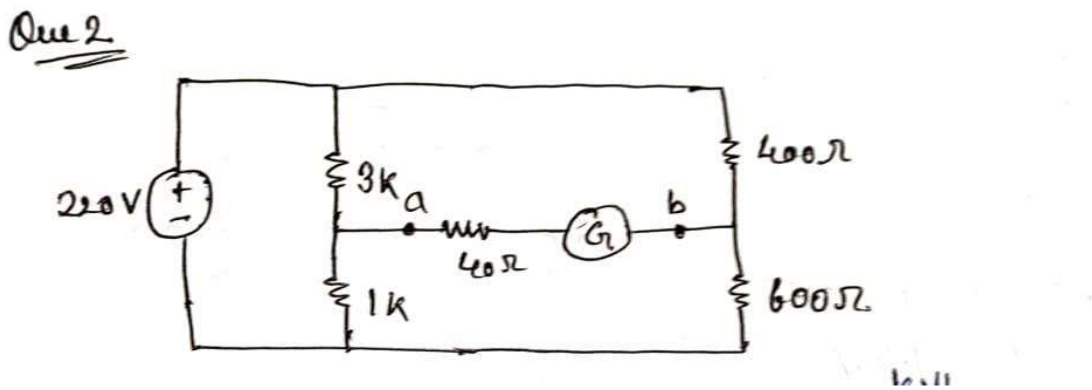
Using Thevenin's theorem find the current Through the galvanometer. (through 40 ohm)

Circuit Diagram:

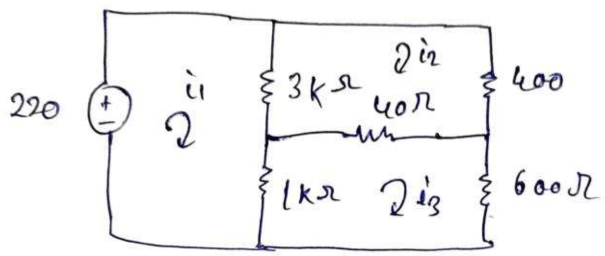


Theoretical Calculation:

Determination of Thevenin's Voltage V_{th}:



Ques. 2



For i_G

$$-220 + 4000 i_1 - 3000 i_2 - 1000 i_3 = 0$$

$$3440 i_2 - 40 i_3 - 3000 i_1 = 0$$

$$1640 i_3 - 40 i_2 - 1000 i_1 = 0$$

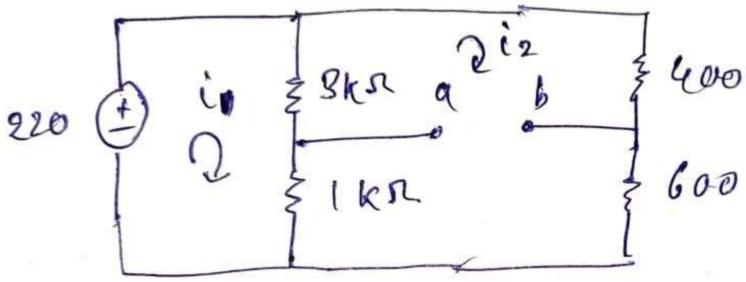
$i_1 = 0.301\ A$

$i_2 = 0.264\ A$ $\qquad i_G = 0.264 - 0.19 = 0.074\ A$

$i_3 = 0.19\ A$

$$\boxed{i_G = 0.074\ A}$$

For V_{th}.



$$220 = 4000\, i_1 - 4000\, i_2 \quad \text{— (1)}$$

$$i_1 = 0.275, \quad i_2 = 0.22$$

$$5000\, i_2 - 4000\, i_1 = 0 \quad \text{— (2)}$$

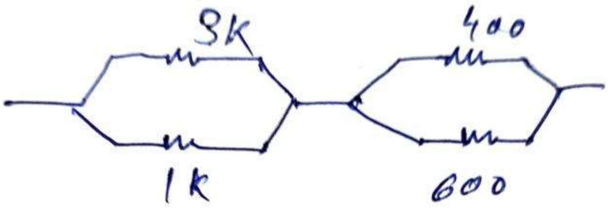
$$V_{th} = -3000\,(0.275 - 0.22) + 400\,(0.22)$$

$$= -165 + 88$$

$$\boxed{V_{th} = 77V}$$

Determination of Thevenin's Resistance R_{th}:

For R_{th}.

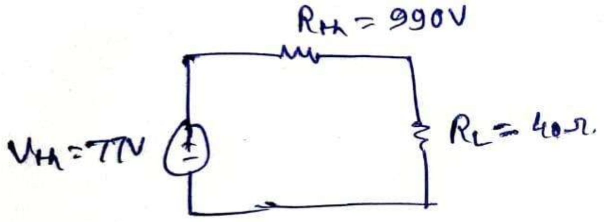


$$R_{th} = \frac{3}{4}k + 240$$

$$\boxed{R_{th} = 990\,\Omega}$$

Thevenin's Equivalent Circuit and Load Current I_L Calculation:

Thevenin's Equivalent Circuit

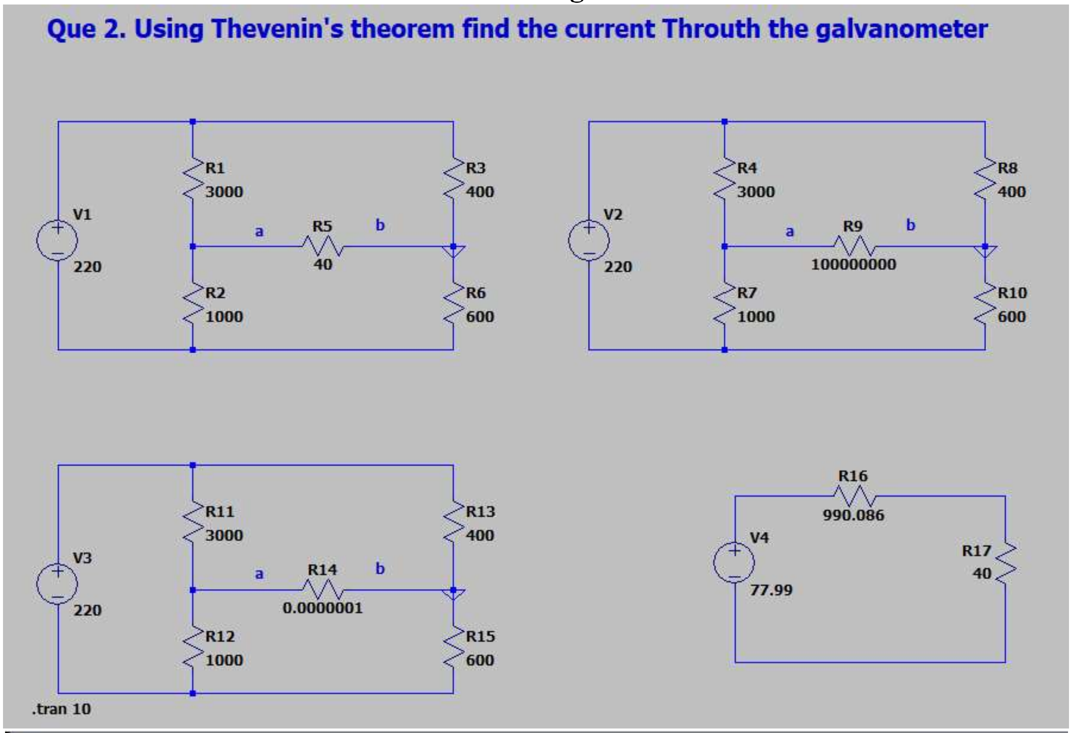


$$I_L = \frac{V_{th}}{R_L + R_{th}} = \frac{77}{990+40}$$

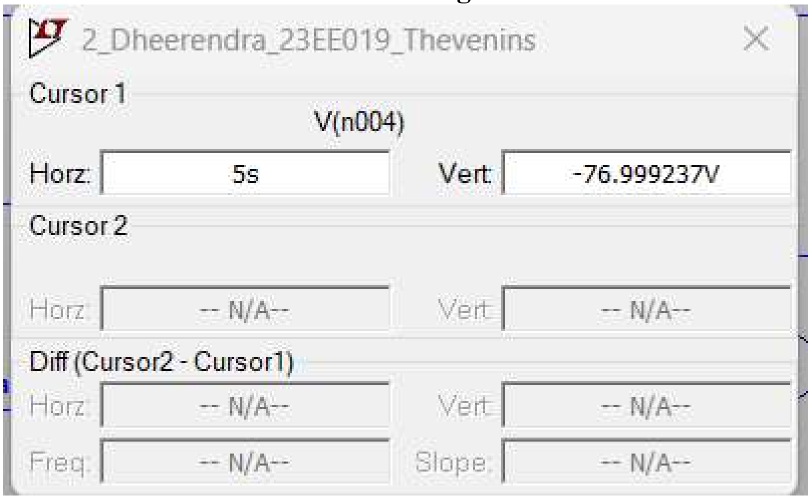
$$\boxed{I_L = 0.07476\ A}$$

Verification by Simulation:

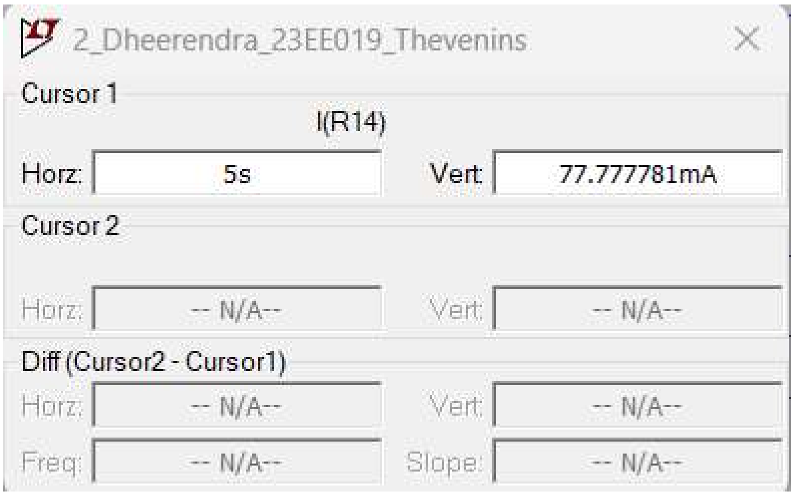
Circuit Diagram:



Determination of Thevenin's Voltage: -77.999237V



Determination of Thevenin's Resistance: 989.98978 ohm



Vth = 76.999237 V

Isc = 77.777781 mA

Rth = Vth/Isc = 989.98978 ohm

Determination of Load Current from Thevenin's Equivalent Circuit:

Observation Table:

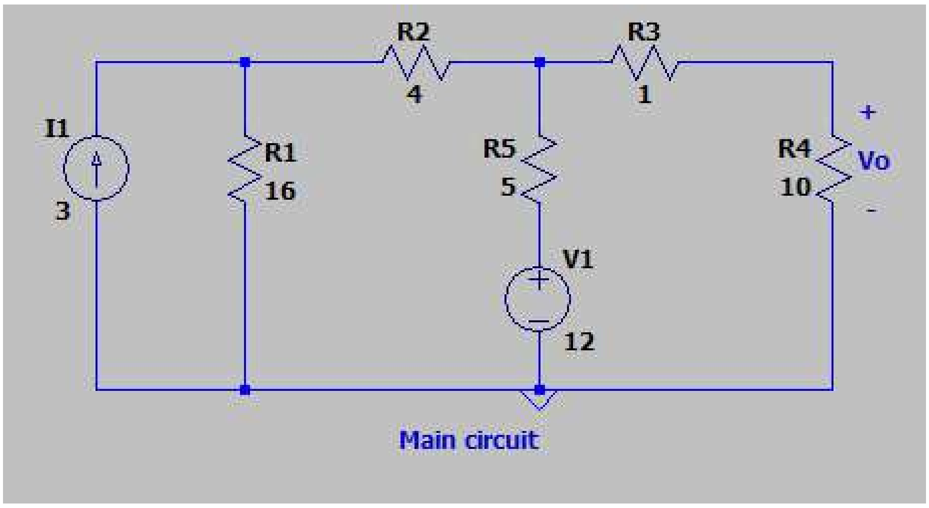
S. No.	V_{th} (V)	R_{th} (Ω)	Current through Load Resistance I_L (A)	
			Practical Value	Theoretical Value

2 (Theoretical)	77	990	Main Circuit	0.074757278	Main Circuit	0.074
2 (Practical)	76.999237	989.98978	Thevenin's Circuit	0.075712122	Thevenin's Circuit	0.07476

Question 3

Using Thevenin's theorem find Vo in the circuit

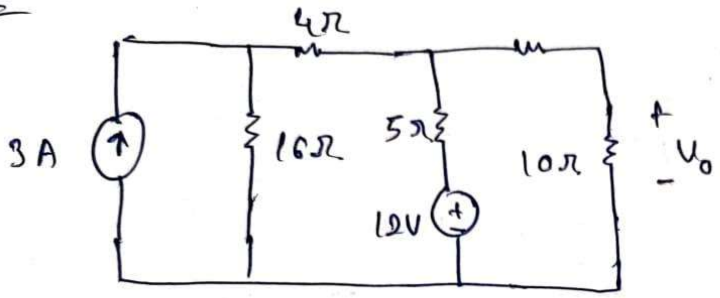
Circuit Diagram:



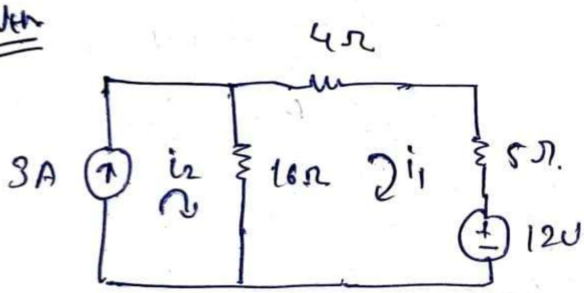
Theoretical Calculation:

Determination of Thevenin's Voltage V_{th}:

Que 3



For Vth



$$i_2 = 3A$$

$$25i_1 - 16(3) = -12$$

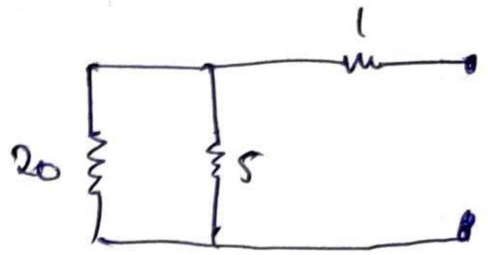
$$i_1 = 1.44A$$

$$V_{th} = 5 \times 1.44 + 12$$

$$\boxed{V_{th} = 19.2V}$$

Que

For R_{th}

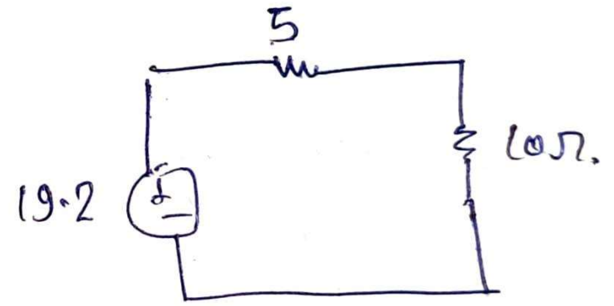


$\boxed{R_{th} = 5\Omega.}$

Determination of Thevenin's Resistance R_{th}:

Thevenin's Equivalent Circuit and Load Current I_L Calculation:

thevenin equivalent Circuit

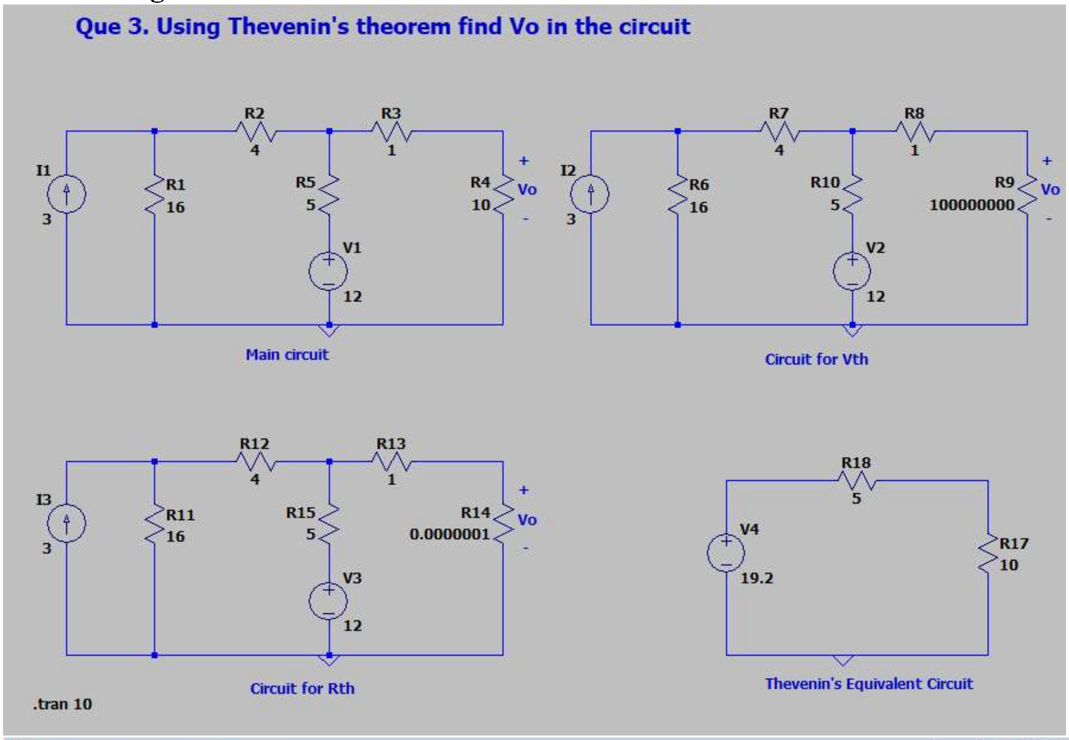


$$i_{10\Omega} = \frac{19.2}{15}$$

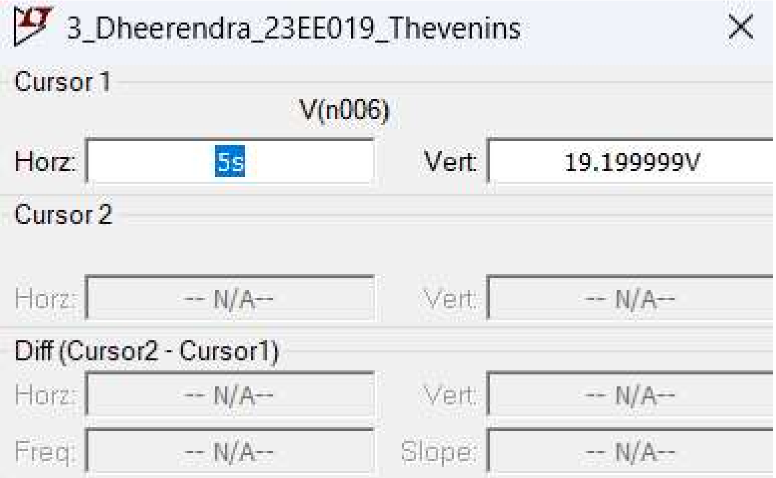
$$\boxed{i = 1.28\,A}$$

Verification by Simulation:

Circuit Diagram:



Determination of Thevenin’s Voltage:



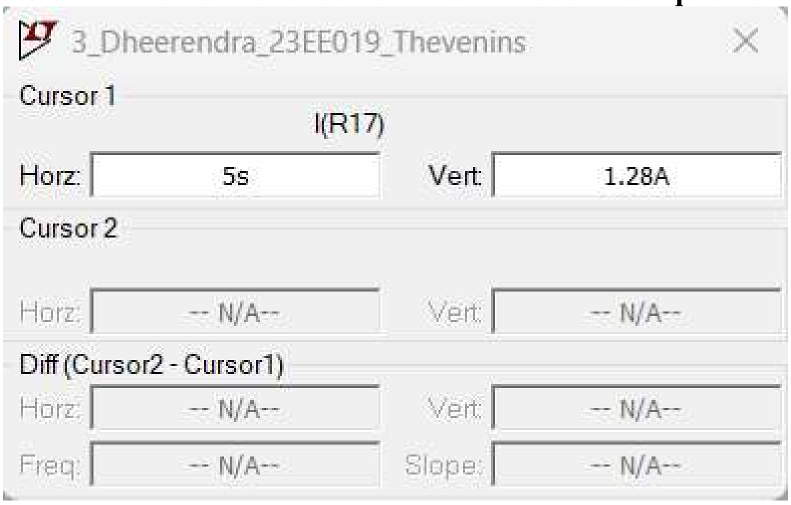
Determination of Thevenin’s Resistance:

Vth = 19.199999 V

Isc= 3.839999 A

Rth =Vth/Isc = 5 ohm

Determination of Load Current from Thevenin’s Equivalent Circuit:



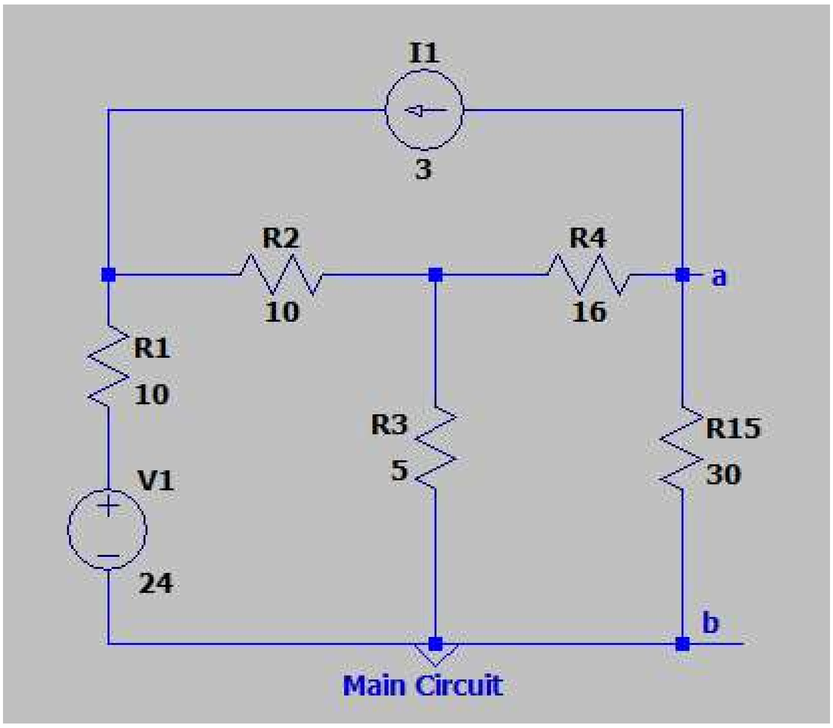
Observation Table:

S. No.	V_{th} (V)	R_{th} (Ω)	Current through Load Resistance I_L (A)			
			Practical Value		Theoretical Value	
3. (Theoretical)	19.2	5	Main Circuit	1.28	Main Circuit	1.28
3. (Practical)	19.199999	5	Thevenin's Circuit	1.28	Thevenin's Circuit	1.28

Question 4

Obtain the Thevenin's equivalent at terminals a-b of the circuit shown and determine the current through R=30 ohm connected between terminal a-b.

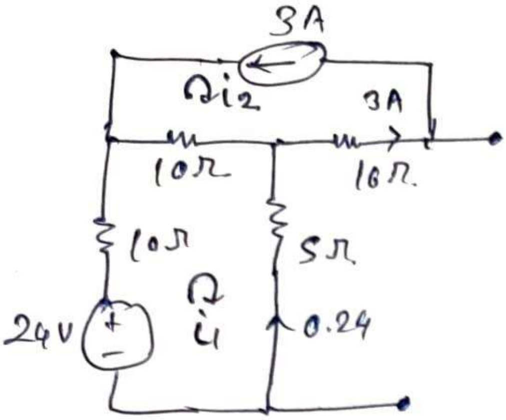
Circuit Diagram:



Theoretical Calculation:

Determination of Thevenin's Voltage V_{th}:

Que 4.



for V_{th}

$$i_2 = -3\ A$$

$$24 = 25 i_1 - 10 i_2$$

$$i_1 = -0.24\ A$$

$$i_2 = -3\ A$$

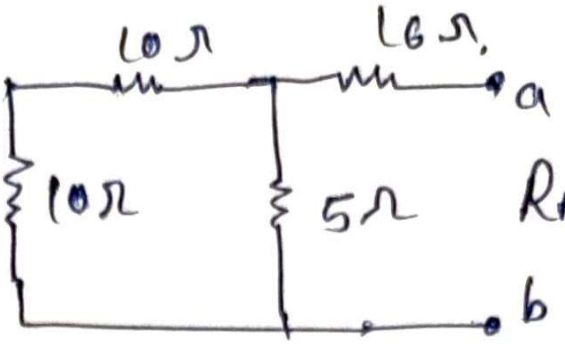
$$V_{th} = 0.24 \times 5 + 3 \times 16$$

$$\boxed{V_{th} = 49.2V}$$

Determination of Thevenin's Resistance R_{th}:

For R_{th}.

R_{th}



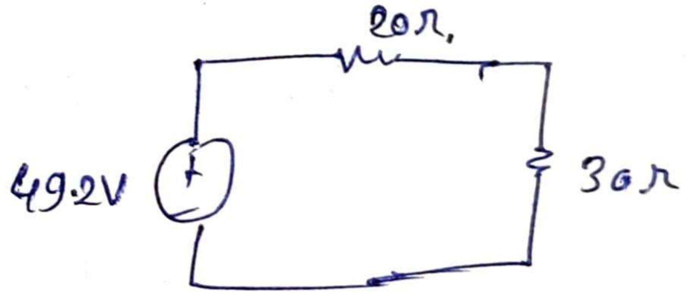
4+16

$$\boxed{R_{th} = 20\Omega}$$

Thevenin's Equivalent Circuit and Load Current I_L Calculation:

thevenins Equivalent Circuit

$$i_{30\Omega} = \frac{49.2}{50} = 0.984A$$



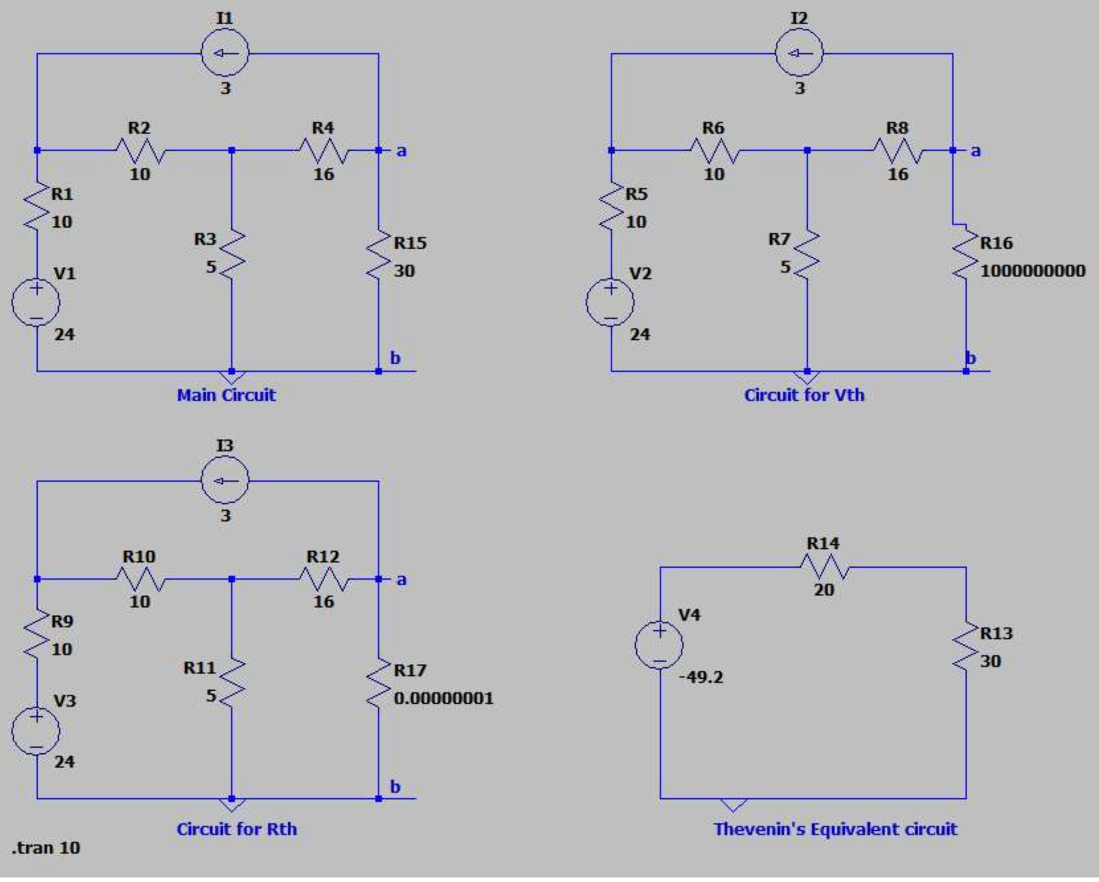
$$\boxed{i_{30\Omega} = 0.984}$$

Verification by Simulation:

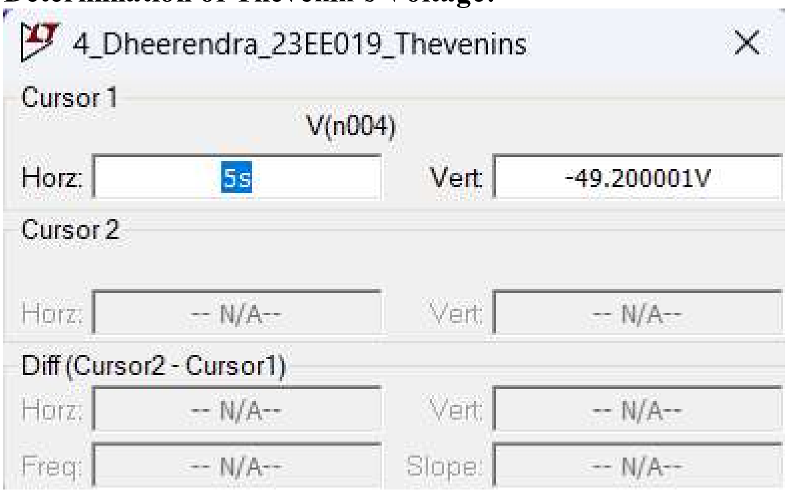
Circuit Diagram

:

Que 4. Obtain the Thevenin's equivalent at terminals a-b of the circuit shown and determine the current through R=30 ohm connected between terminal a-b



Determination of Thevenin's Voltage:



Determination of Thevenin's Resistance:

Vth = -49.2V

Isc = -2.46 A

Rth = 20 ohm

Determination of Load Current from Thevenin's Equivalent Circuit:

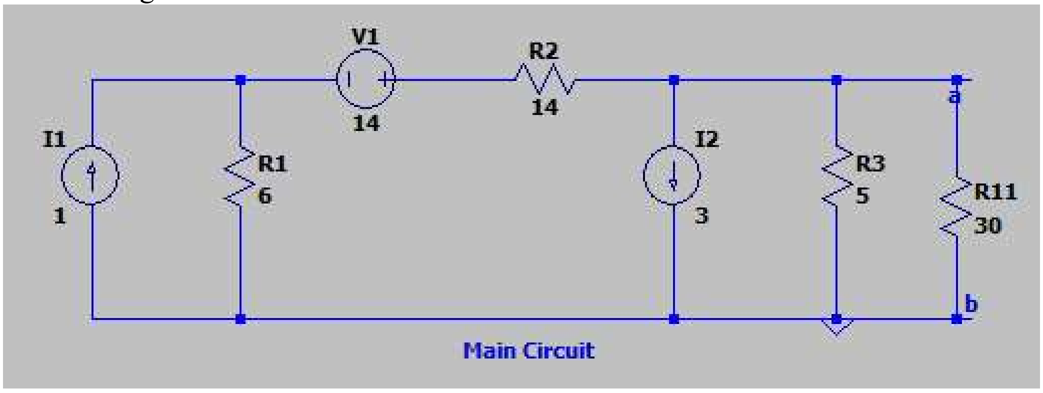
Observation Table:

S. No.	V_{th} (V)	R_{th} (Ω)	Current through Load Resistance I_L (A)			
			Practical Value		Theoretical Value	
4. (Theoretical)	49.2	20	Main Circuit	0.98400003	Main Circuit	0.984
4. (Practical)	49.2	20	Thevenin's Circuit	0.98400003	Thevenin's Circuit	0.984

Question 5.

Obtain the Thevenin's equivalent at terminals a-b of the circuit shown and determine the current through R=30 ohm connected between terminal a-b.

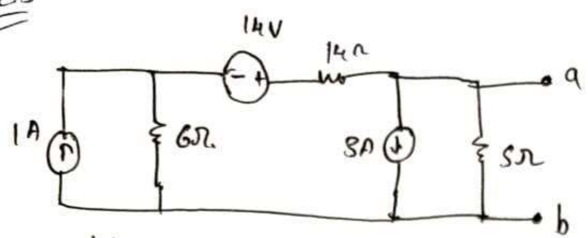
Circuit Diagram:



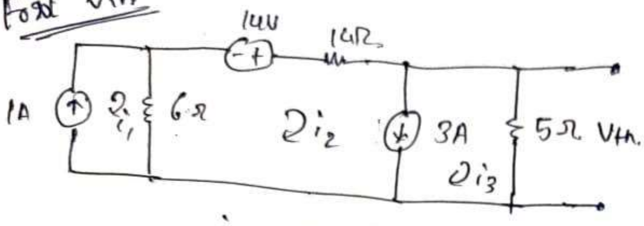
Theoretical Calculation:

Determination of Thevenin's Voltage V_{th}:

Que.5



For Vth



$V_{th} = 7.999\,V$

$I_{th} = 1.999\,A$

$R_{th} = 4.00015$

$i_1 = 1A$

$i_2 = i_3 + 3$

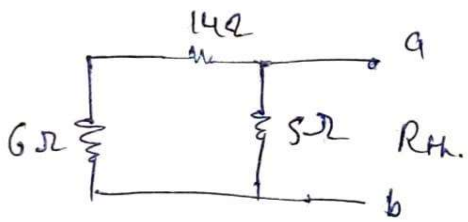
$= 6i_1 + 20i_2 + 5i_3 - 14 = 0$

$i_3 = -1.6A$

$\boxed{V_{th} = 8V}$

Determination of Thevenin's Resistance R_{th}:

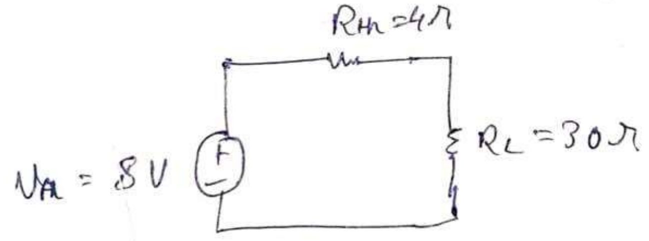
For Rth.



$\boxed{R_{th} = 4\Omega}$

Thevenin's Equivalent Circuit and Load Current I_L Calculation:

thevenins Equivalent Circuit

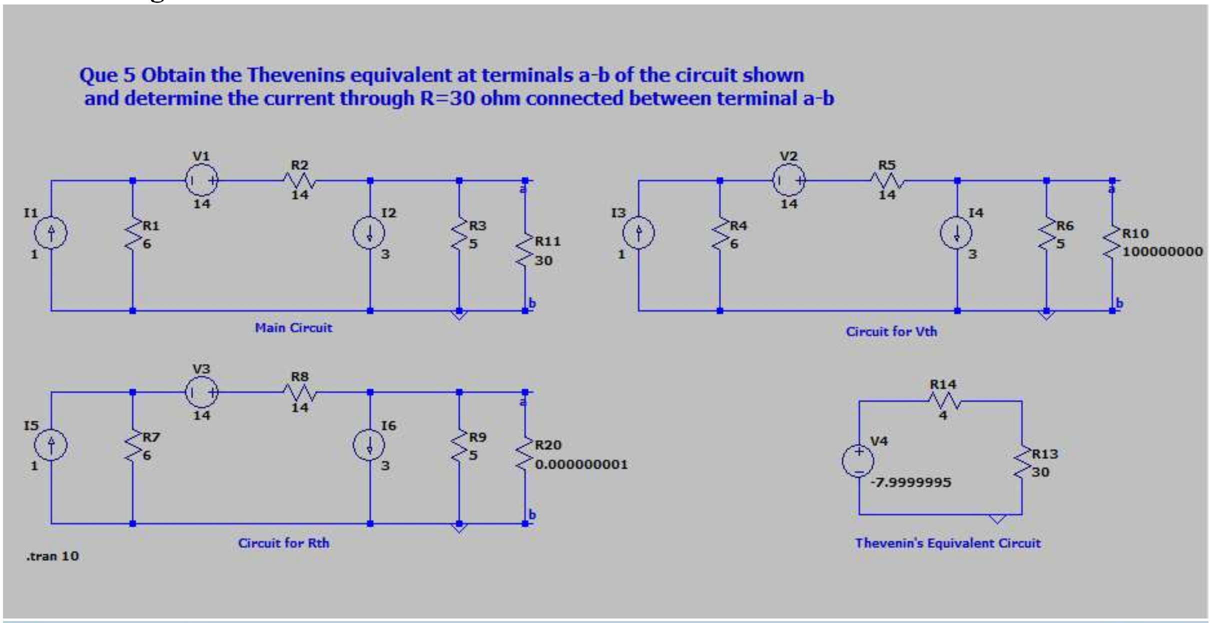


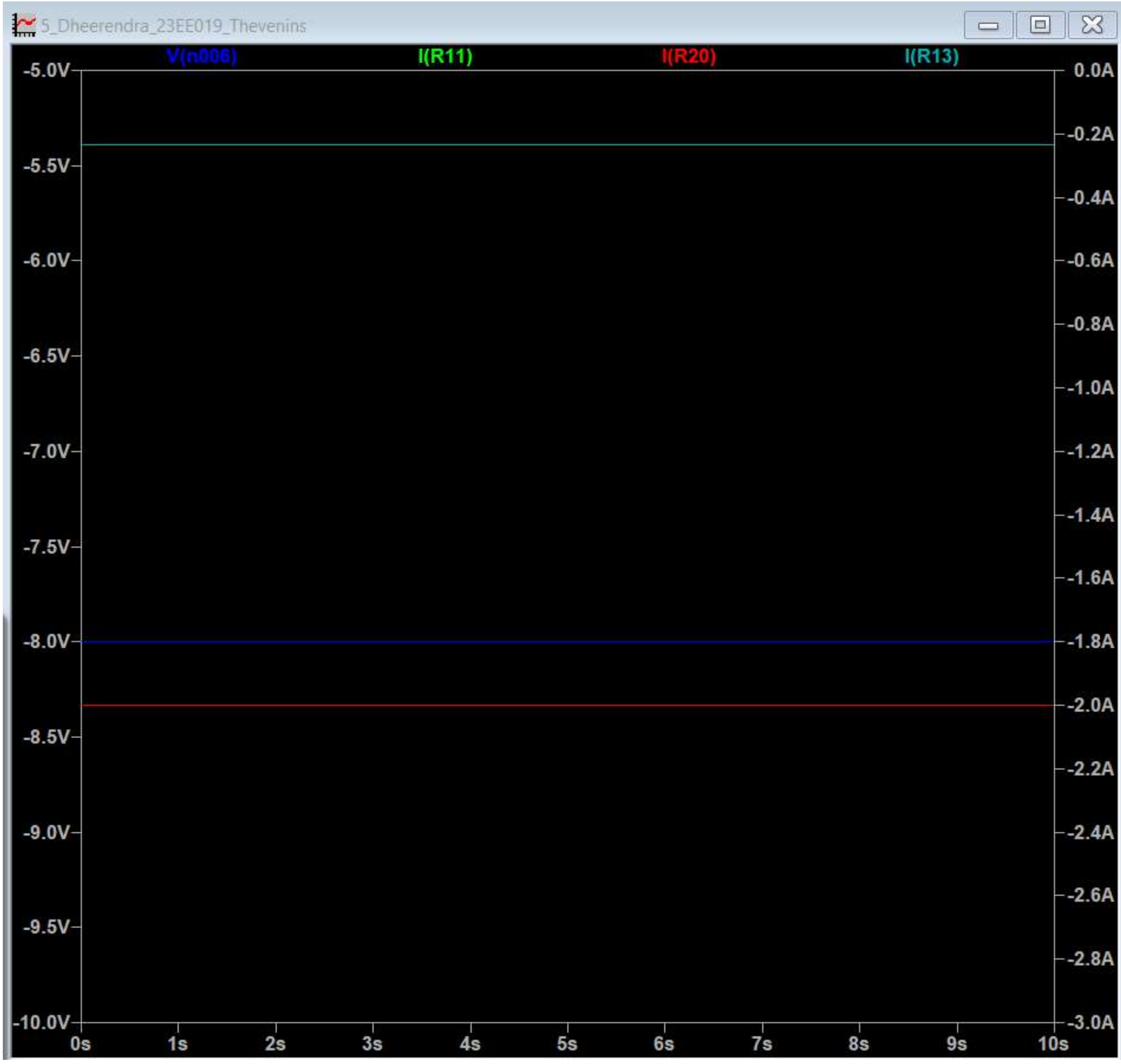
$$I_{30\Omega} = \frac{8}{34} = 0.23529A$$

$$\boxed{I_{30\Omega} = 0.23529A}$$

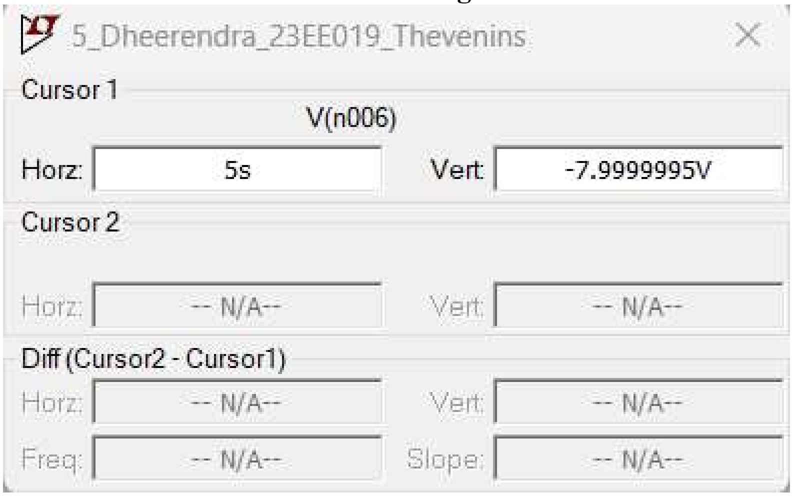
Verification by Simulation:

Circuit Diagram:





Determination of Thevenin's Voltage:



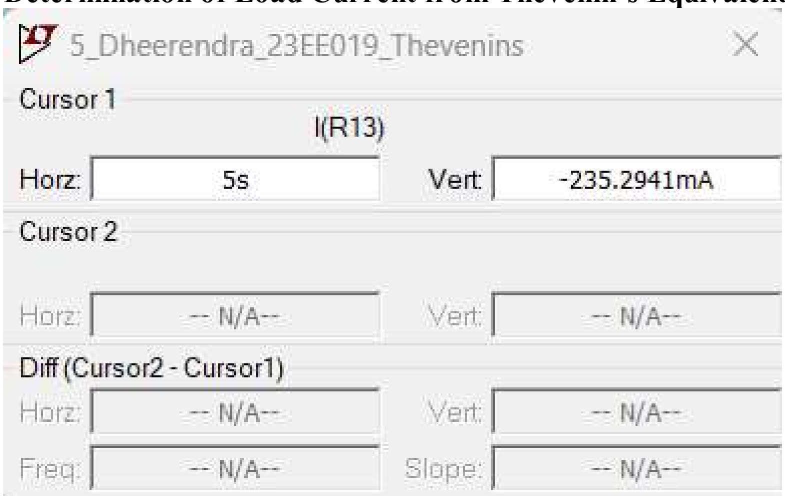
Determination of Thevenin's Resistance:

Vth = -7.9999995 V

Isc = -2 A

Rth = Vth/Isc = 4 ohm

Determination of Load Current from Thevenin's Equivalent Circuit:



Observation Table:

S. No.	V_{th} (V)	R_{th} (Ω)	Current through Load Resistance I_L (A)			
			Practical Value		Theoretical Value	
5. (Theoretical)	8	4	Main Circuit	0.23529	Main Circuit	0.235294
5. (Practical)	7.9999	4	Thevenin's Circuit	0.23529	Thevenin's Circuit	0.235294

Result:

As we see in the above 5 Question that theoretical values is approximately equal to the practical Values, Hence Thevenin's Theorem is verified.